

## Industrial Internship Report on

# Employee Attrition Prediction Using Machine Learning

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### *Executive Summary*

This report documents the work completed during the Machine Learning internship, focused on predicting employee attrition using the IBM HR Analytics Employee Attrition dataset. The project involved exploratory data analysis, feature engineering, and training a Random Forest classifier to identify employees at risk of leaving the organisation.

The trained model achieved an overall accuracy of 83.3% on the held-out test set, and feature importance analysis revealed monthly income, age, total working years, and overtime status as the strongest predictors of attrition.

This internship provided practical, hands-on exposure to the complete data science workflow — from raw data to a deployable predictive model — and strengthened my understanding of how machine learning can support real HR decision-making.

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## 1. Preface

This report summarises the work carried out over the course of the internship, during which I worked on building a machine learning model to predict employee attrition — i.e., whether an employee is likely to leave the organisation — using historical HR data.

Internships like this one are valuable for bridging the gap between classroom learning and industry practice. They provide an opportunity to work with real, messy datasets, make design decisions under time constraints, and see a project through from data ingestion to a working, evaluated model.

The project undertaken was the design and implementation of an Employee Attrition Prediction system. The work was planned in stages: data understanding and cleaning, exploratory data analysis (EDA), feature engineering, model training, evaluation, and interpretation of results through feature importance analysis.

Over the course of this internship, I learned how to systematically approach a classification problem: starting with understanding the business question (why do employees leave?), moving through data preparation, and finally validating whether a model's predictions are both accurate and interpretable enough to be useful to HR stakeholders.

I would like to thank my mentors and team coordinators for their guidance and feedback throughout this project, and my fellow interns for their collaboration and support.

To juniors and peers undertaking similar projects: spend real time on exploratory data analysis before jumping to modelling — understanding your data's quirks (imbalanced classes, redundant columns, correlated features) pays off far more than tuning hyperparameters blindly.

## 2. Introduction

### 2.1 About the Organisation / Programme

This internship was carried out as part of a structured internship programme focused on applied machine learning and data analysis. The programme is designed to give students exposure to end-to-end data science workflows: from raw, real-world datasets to trained, evaluated, and interpretable models.

The broader curriculum emphasised practical tools used in the industry — Python, pandas, scikit-learn, and standard visualization libraries (matplotlib, seaborn) — along with best practices in reproducibility, version control (GitHub), and clear technical reporting.

### 2.2 Objectives of this Internship

- To gain hands-on experience working with a real-world, industry-style HR dataset.
- To apply the full data science pipeline: cleaning, EDA, feature engineering, modelling, and evaluation.
- To build a classification model capable of predicting employee attrition with reasonable accuracy.
- To interpret model outputs (feature importances) to generate actionable business insight.

- To improve technical documentation and reporting skills for data science projects.

## 2.3 Reference

- [1] IBM HR Analytics Employee Attrition & Performance dataset (WA\_Fn-UseC\_HR-Employee-Attrition.csv).
- [2] scikit-learn documentation — <https://scikit-learn.org/stable/>
- [3] pandas documentation — <https://pandas.pydata.org/docs/>

## 2.4 Glossary

| Term                | Meaning                                                                              |
|---------------------|--------------------------------------------------------------------------------------|
| Attrition           | An employee voluntarily or involuntarily leaving the organisation                    |
| EDA                 | Exploratory Data Analysis — visual/statistical inspection of data before modelling   |
| Feature Engineering | Transforming raw columns into a form usable by a model (encoding, dropping, scaling) |
| Random Forest       | An ensemble of decision trees used for classification/regression                     |
| Feature Importance  | A measure of how much each input variable contributes to model predictions           |
| Confusion Matrix    | A table summarising correct vs incorrect predictions by class                        |

## 3. Problem Statement

Employee attrition — employees leaving an organisation — is costly to businesses in terms of recruitment, onboarding, and lost institutional knowledge. HR departments often lack a systematic, data-driven way to identify which employees are at higher risk of leaving before it happens.

The problem statement for this project was: given historical HR data containing employee demographics, job characteristics, satisfaction scores, and compensation details, build a machine learning model that predicts whether an employee is likely to leave the company (Attrition = Yes/No), and identify the key factors driving that risk.

The dataset used was the IBM HR Analytics Employee Attrition dataset, containing 1,470 employee records across 35 attributes, including Age, MonthlyIncome, JobSatisfaction, OverTime, YearsAtCompany, and the target column, Attrition.

## 4. Existing and Proposed Solution

Traditionally, HR teams rely on exit interviews, manager feedback, and periodic engagement surveys to gauge attrition risk. These approaches are reactive — they identify the reasons for attrition only after an employee has already decided to leave — and are difficult to scale across large workforces.

Existing analytics approaches in the literature typically use logistic regression or single decision trees, which are simple to interpret but often underperform on datasets with class imbalance and mixed

categorical/numerical features, as is the case here (only about 16% of employees in this dataset left the company).

The proposed solution in this project uses a Random Forest classifier, an ensemble method that combines many decision trees to improve predictive accuracy and robustness to overfitting, while still allowing feature importance to be extracted for interpretability. The value addition of this approach is that it gives HR teams both a prediction (who is at risk) and an explanation (why), which is more actionable than a black-box score alone.

## 4.1 Code and Report Submission

Code submission (GitHub link): [add repository link here]

Report submission (GitHub link): [add repository link here]

## 5. Proposed Design / Model

The solution follows a standard supervised machine learning pipeline, moving from raw data to a trained, evaluated, and interpretable model, as illustrated below.

### 5.1 High-Level Workflow

- Stage 1 — Data Loading & Inspection: load the CSV, check shape, data types, and missing values.
- Stage 2 — Exploratory Data Analysis: visualise target distribution, correlations, age distribution, and income vs. attrition.
- Stage 3 — Feature Engineering: drop non-informative columns (EmployeeCount, Over18, StandardHours, EmployeeNumber), encode the target and categorical variables (one-hot encoding).
- Stage 4 — Train/Test Split: 80/20 split, stratified on the target to preserve class balance.
- Stage 5 — Model Training: Random Forest Classifier (scikit-learn, random\_state=42).
- Stage 6 — Evaluation: accuracy, precision/recall/F1, confusion matrix.
- Stage 7 — Interpretation: feature importance ranking to identify key attrition drivers.
- Stage 8 — Persistence: trained model serialised with joblib for reuse/deployment.

### 5.2 High Level Diagram

Raw HR Data (CSV) → Data Cleaning & Feature Engineering → Encoded Feature Matrix → Train/Test Split → Random Forest Model → Evaluation Metrics → Feature Importance Insights → Saved Model (.pkl).

### 5.3 Low Level Diagram

At a lower level, the pipeline reads the CSV with pandas, drops four non-informative columns (EmployeeCount, Over18, StandardHours, EmployeeNumber), maps the Attrition target to 0/1, and applies `pd.get_dummies()` for one-hot encoding of categorical fields. This produces a 1,470 x 44 feature matrix, which is split 80/20 (stratified) before being passed to scikit-learn's `RandomForestClassifier` for training and inference.

## 5.4 Interfaces

The current implementation is a self-contained Python/Colab notebook (ML.py) with no external interfaces. The natural next interface is a lightweight REST API or dashboard that accepts an employee record as input and returns an attrition-risk score plus the top contributing features, described further under Future Work Scope.

## 5.5 Exploratory Data Analysis

The dataset contains 1,470 rows and 35 columns with no missing values, so no imputation was required. The target variable, Attrition, is imbalanced: roughly 16% of employees are labelled as having left the company, versus 84% who stayed.

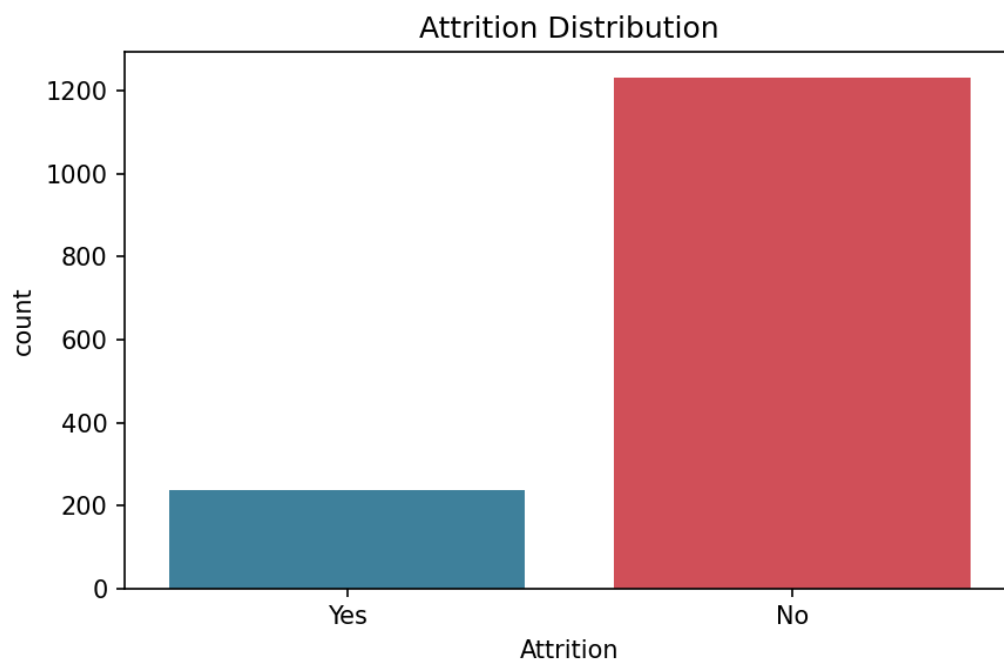


Figure 1: Attrition class distribution — showing the imbalance between employees who stayed vs. left.

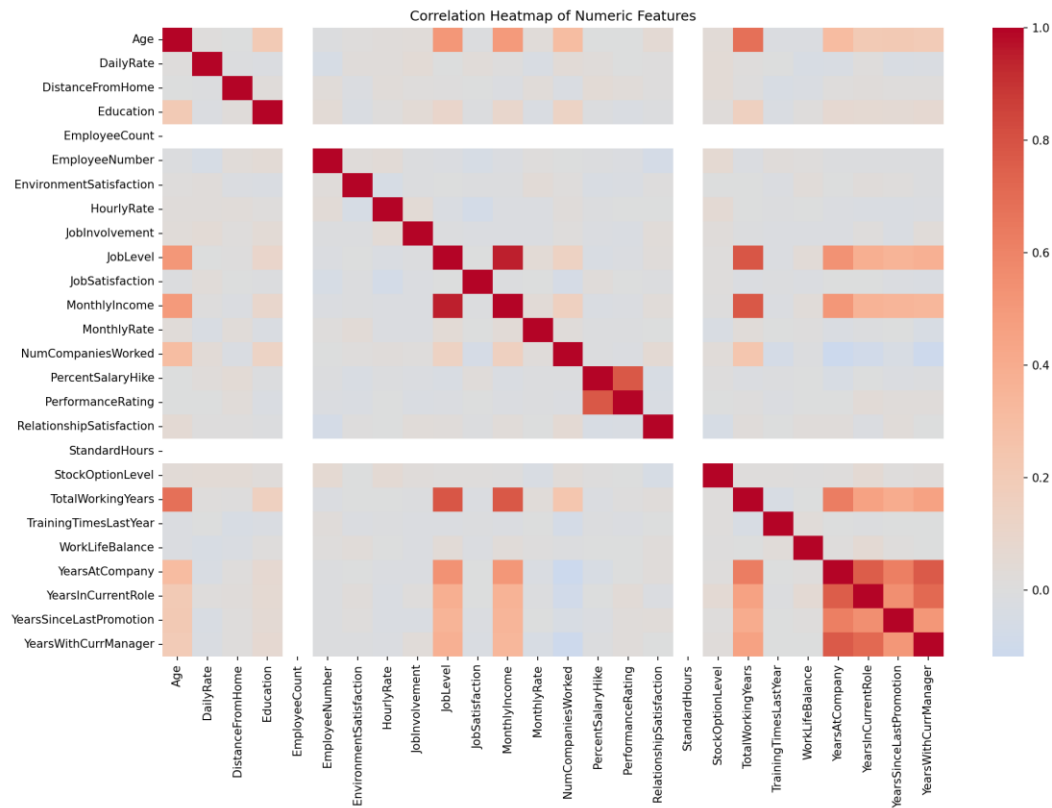


Figure 2: Correlation heatmap across all numeric features, used to spot redundant or highly correlated variables.

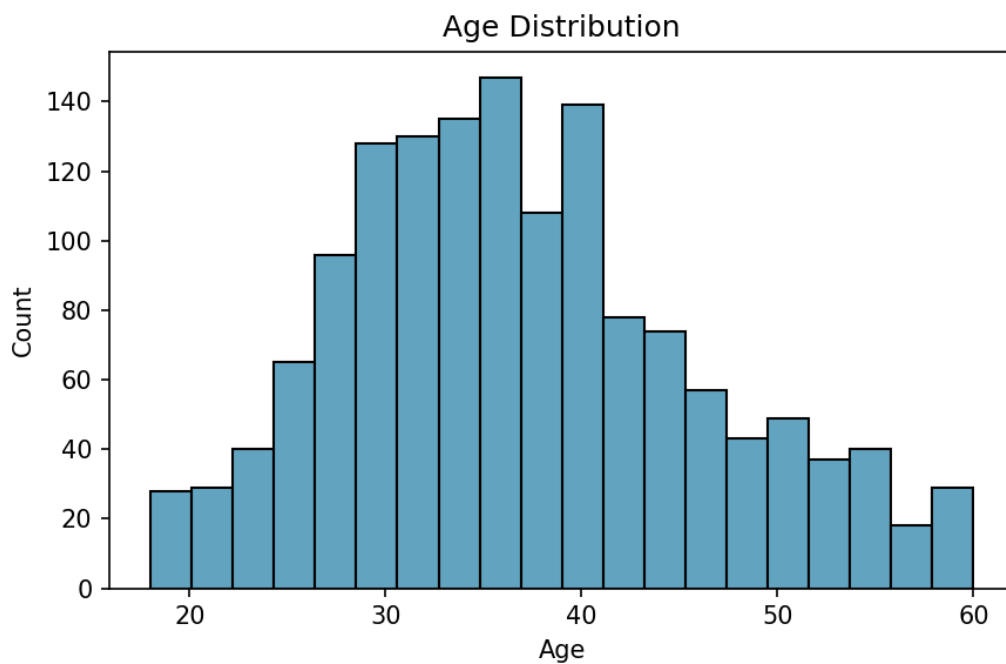


Figure 3: Age distribution of employees in the dataset, mostly concentrated between 28–40 years.

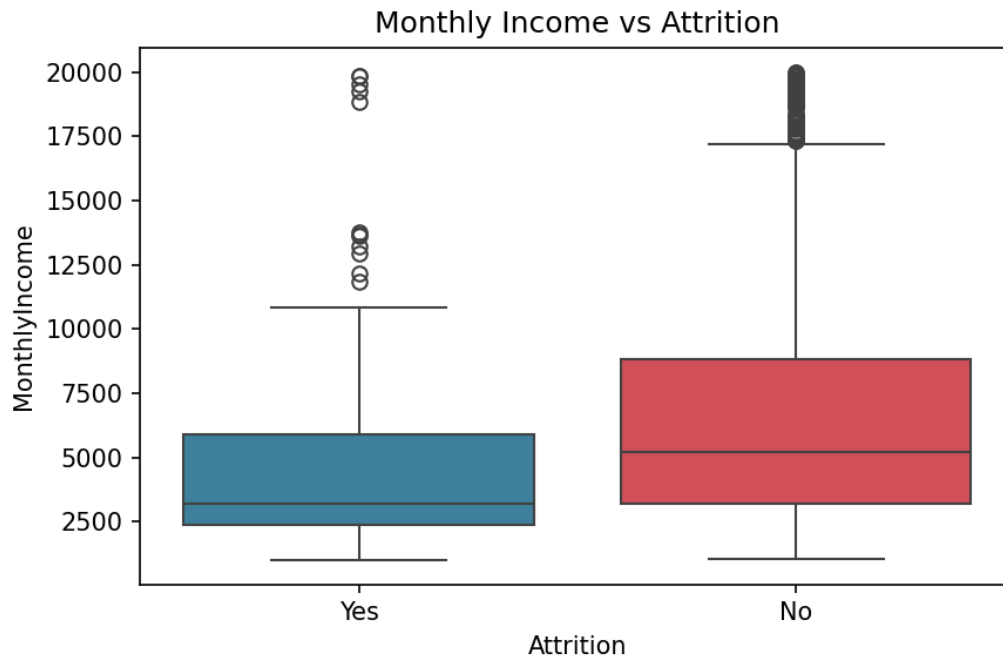


Figure 4: Monthly income compared across employees who left vs. stayed — employees who left tend to have lower median income.

## 5.6 Feature Engineering

Four columns were dropped prior to modelling as they carried no predictive signal: EmployeeCount and StandardHours are constant across all rows, Over18 is constant ('Y' for every employee), and EmployeeNumber is a unique identifier rather than a feature. The target column, Attrition, was mapped from Yes/No to 1/0, and all remaining categorical columns (e.g., Department, JobRole, MaritalStatus, OverTime) were one-hot encoded, expanding the feature set from 31 to 44 predictor columns.

## 6. Performance Test

The key constraint for this project was predictive performance on an imbalanced target — a model that simply predicts 'No attrition' for every employee would already score roughly 84% accuracy while being practically useless to HR. The evaluation therefore looked beyond raw accuracy to precision, recall, and F1-score for the minority ('Attrition = Yes') class.

### 6.1 Test Plan / Test Cases

The dataset was split 80/20 into training and test sets (1,176 training rows, 294 test rows), stratified by the target so that the same ~16% attrition ratio is preserved in both splits. The Random Forest classifier was trained on the training split with 100 estimators and default depth, and default hyperparameters otherwise (random\_state=42 for reproducibility).



## 6.2 Test Procedure

The trained model was used to predict attrition labels on the held-out test set. Predictions were compared against the true labels using `accuracy_score`, `classification_report`, and `confusion_matrix` from `scikit-learn`.

## 6.3 Performance Outcome

The model achieved an overall accuracy of 83.3% on the test set. However, class-level performance highlights the impact of the imbalance: the model performs strongly on identifying employees who stay (precision 0.85, recall 0.97) but weakly on identifying employees who actually leave (precision 0.42, recall 0.11), correctly catching only 5 of 47 true attrition cases in the test set.

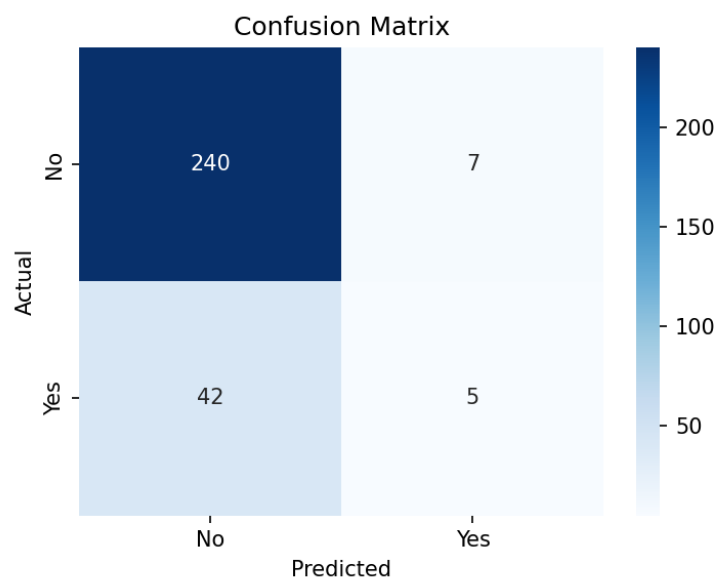


Figure 5: Confusion matrix on the test set (294 employees) — the model misses most true attrition cases (42 false negatives).

This is a common and important failure mode for imbalanced classification problems: high overall accuracy can mask poor performance on the minority class that matters most to the business. Recommendations for handling this in future iterations include class-weighting or resampling techniques (e.g., SMOTE), threshold tuning to trade off precision/recall, and collecting more examples of attrition cases if possible.

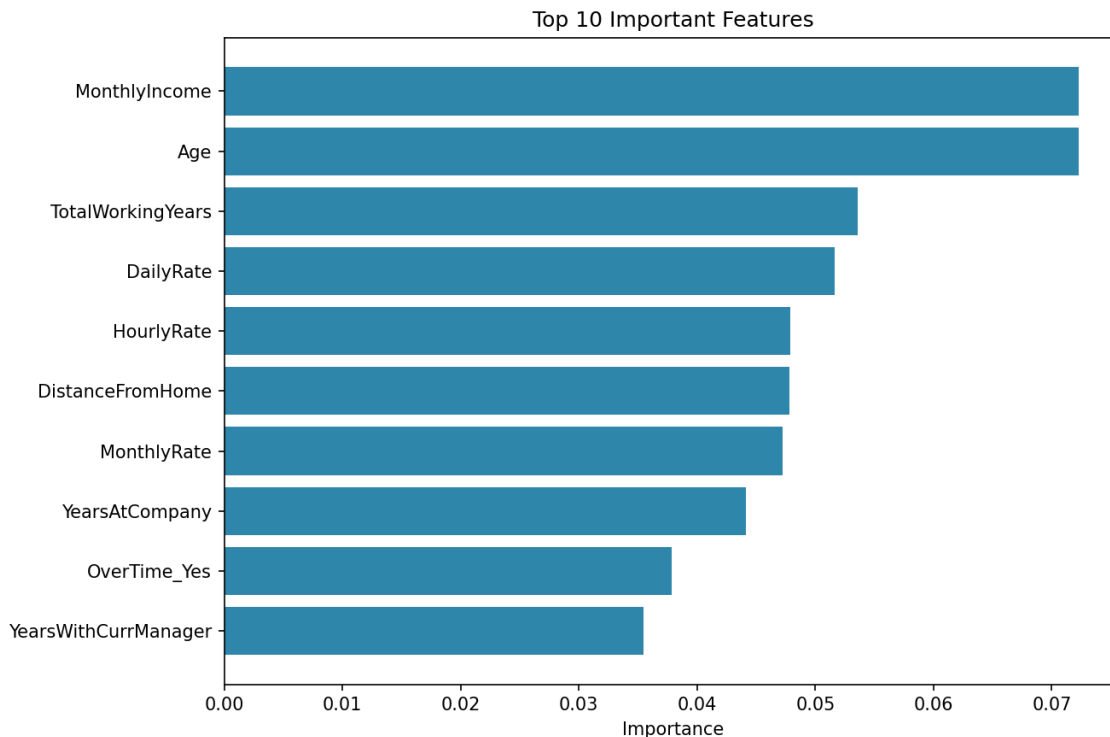


Figure 6: Top 10 features by importance — Monthly Income, Age, and Total Working Years were the strongest predictors.

The feature importance results are business-relevant: compensation (MonthlyIncome, DailyRate, HourlyRate, MonthlyRate), tenure/experience (Age, TotalWorkingYears, YearsAtCompany), commute burden (DistanceFromHome), and OverTime status all rank among the top predictors — aligning with the intuitive HR narrative that under-compensated, overworked, or less-tenured employees are at greater flight risk.

## 7. My Learnings

This internship gave me practical, end-to-end experience with a real machine learning workflow, rather than a toy or pre-cleaned dataset. Key takeaways include:

- The importance of exploratory data analysis before modelling — visualising distributions and correlations surfaced issues (constant columns, class imbalance) that would otherwise have gone unnoticed.
- Why accuracy alone is a misleading metric on imbalanced datasets, and why precision, recall, and the confusion matrix give a fuller picture.
- How feature importance from tree-based models can turn a black-box prediction into an interpretable, business-relevant insight.
- Practical skills in pandas for data cleaning and encoding, and scikit-learn for model training, evaluation, and persistence (joblib).

- The value of clear, reproducible code and documentation when handing off a project or reporting results to non-technical stakeholders.

These skills — structured problem framing, rigorous evaluation, and clear communication of results — will directly support my growth in future data science and machine learning roles.

## 8. Future Work Scope

- Address class imbalance explicitly using techniques such as SMOTE, class-weighted loss, or undersampling the majority class.
- Experiment with alternative models (Gradient Boosting, XGBoost, Logistic Regression with class weights) and compare performance.
- Perform hyperparameter tuning (grid search / randomized search) on the Random Forest to improve minority-class recall.
- Deploy the trained model behind a simple API or dashboard so HR teams can query attrition risk for individual employees.
- Incorporate additional data sources (engagement survey scores, manager feedback) if available, to improve recall on true attrition cases.